19	A container contains a volume of 120 cm^3 of water at 60 °C. 30 cm^3 of water at temperature of 20 °C is added to the container. Assume that no heat escapes from the container and the density of water remains constant. What is the temperature of the combined volume of water when thermal equilibrium is reached?							
	A	26 °C	**B**	28 °C	**C**	45 °C	**D**	52 °C
L2	**Answer: D** Let θ be the equilibrium temperature, ρ be the density of water, and c be the specific heat capacity of water.							

	By conservation of energy, Heat loss by the 120 cm^3 water = Heat gained by the 30 cm^3 water $Mc(60 - \theta) = mc(\theta - 20)$ $120\rho c(60 - \theta) = 30\rho c(\theta - 20)$ $4(60 - \theta) = (\theta - 20)$ $240 - 4\theta = \theta - 20$ $5\theta = 260$ $\theta = 52°$